

THE IDEAL OS

This is indeed the one OS to rule them all. If it were to be built that is...

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The last couple of years have seen some massively disruptive changes in computing, with several new technologies successfully challenging existing ones. This was particularly noticeable in the mobile world, where the shift towards computing on smartphones brought with it the need for specialised mobile platforms.

The different Operating Systems out there offer a lot of diversity in terms of their philosophy, systems design, user interface and third-party application support. Using a mix of existing technologies and some extrapolative reasoning, we've tried to come up with our version of what the ideal OS will be like.

The Ideal Operating System Will Not Exist

The Ideal Operating System (or at least, our version of it) will simply not exist. No we're not pessimists. In our vision the ideal OS will quietly disappear into the background, providing all necessary functions silently and without any outwardly signs. Allow us to elaborate.

Imagine a world, where every single personal computing device, is nothing but a terminal connected to a server. (In some sense, this is true for all internet-enabled devices even today). The device you hold in your hand or the one that's at your desk, will

be nothing but a display, letting you access your apps and multimedia, all stored in the cloud. You sign into a device (any device, it doesn't matter) with your credentials, and it automatically pulls all your data, applications and media, hell, even the OS itself will be streamed to you via this network. Think of it as accessing web mail – when you buy a device and load up the webmail URL in a browser, you know for a fact that your email is going to look exactly the same as on every other device, without having to worry about the hardware manufacturer.

"But this already exists" you might say. It does, to an extent. Windows 8 syncs your device's settings and applications to the cloud via SkyDrive, and it's there for you when you log in. But you can only access this feature from a Windows 8 desktop, or a phone running Windows Phone 8 (yes, all 3 units that've been sold so far). A similar feature exists with the Chromebook, but when was the last time you saw someone using one? (Hint: Never).



Imagine a hardware agnostic OS

Separating the Hardware from the Software

What we'd like to see, is a completely hardware agnostic Operating System (or a generation of Operating Systems, from competing companies) that can be made to run on any device, because after all a device is just a means to connect to a server. You will be able to buy potentially any hardware device off the shelf with specifications as per your need, and use it with any OS of your choice. The device will contain just the bare minimum in terms of software – only the system software and drivers required to interface the various hardware components (display, network stack, power management) with each other. This is analogous to our current BIOS software, but much more sophisticated (not hard to imagine when you look at some of the UEFI BIOS versions out there) allowing any OS to be installed on top of it. Except in this case, "installed" is a term that refers to the act signing in to an account which pulls in all your data for you.

Head in the cloud...

In order to make these devices ubiquitous and low-cost, they will offload all hardware intensive tasks onto the server. All important processing or GPU operations will be carried out by the server itself, and only the output will be streamed back to you. (Think www.onlive.com) In other words, your device's hardware won't be a limiting factor in its performance. You'll be able to enjoy high-res games, HD movies, hell, even transcode audio and video from your portable device, knowing that all the huffing and puffing is being managed remotely. Furthermore, all media content like movies, TV shows